FinFET 自熱 -> 飽和區 ID-VG (Ion) 下降 % 文獻目錄

矽 FinFET（含相鄰 nanosheet/GAA）self-heating 對飽和區汲極電流影響之量化文獻總表 | 共 121 篇、去重、對抗式查證（0 篇判定捏造）| IEEE 89 篇、非 IEEE 32 篇 | 2026-07-04

一、直接結論：飽和區 Ion 下降多少 %

標準矽 FinFET、VDD~0.7 V、TA=300 K、電熱 vs 等溫，逐字確認的數字落在 **~7-11%**（與判準帶 bulk 3-12%、SOI 8-17% 一致）。特殊結構（DMG 通道工程、高應力、±1.5 V 高過驅）會更高，屬離群值。核心逐字錨點：

• **14nm FinFET** n **7.26% / p 8.91%** @VDS=0.7V 飽和（Physica Scripta 100, 035937, 2025, 10.1088/1402-4896/adb2f1）← 最貼你的情境

• **3nm bulk FinFET** n **10.6% / p 21.6%** @VG=VD=0.7V（Sci. Rep. 16, 16839, 2026；p 值含應力糾纏，非硬上限）

• 多鰭 SOI FinFET **n >10% / p >7%** @±1.5V（Nam, SST 29, 115021, 2014）；bulk FinFET **7.8%**（Myeong, TED 2019, STI 研究）

判準：飽和 Ion 差 >20%（排除 DMG/JL 特例）代表 SHE 過強->查 thermode 是否過絕熱；<1% 代表 SHE 被邊界條件抹掉->查 thermode 是否貼太近通道。交叉檢核比值 ΔIon% ÷ ΔT ~ 0.10-0.20 %/K。

二、電熱 vs 等溫（electrothermal-vs-isothermal）— 矽 FinFET（16 篇）

這是「模擬中 SHE 效應」的正確定義：同一 deck 開/關晶格加熱之差。標「圖層級」者原文有曲線但未印出單一 %，引用數字前需自讀原圖。

論文 / IEEE / 年 / DOI	元件	飽和 Ion 下降 % / 關鍵數字	偏壓 / ΔT	查證
Impact of Self-Heating on Negative-Capacitance FinFET: Device-Circuit ... (IEEE, 2021) DOI: 10.1109/ted.2021.3059180	14-nm NC-FinFET 與標準 14-nm FinFET 基線	not accessed(全文付費牆,摘要無 %;找不到開放 PDF)	bias:not accessed | dT:not accessed(摘要無溫度數值)	plausible_unveri
Transistor Self-Heating: The Rising Challenge for Semiconductor Testin... (IEEE, 2021) DOI: 10.1109/vts50974.2021.9441002	n/p 型 FinFET(TCAD 以 Intel 14nm 量測數據校準)	figure-level;已證實摘要確無任何 Ion % 數值	bias:摘要未載 | dT:摘要無數值(僅述通道內侷限熱)	confirmed
A Junctionless Accumulation Mode NC-FinFET Gate Underlap Design for Im... (IEEE, 2020) DOI: 10.1109/ted.2020.2997848	7-nm-node junctionless accumulation-mode negative-capacita...	no % printed in abstract: Ion/Ioff ratio and peak gm reported 'with and without self-heating effects' at figure level; f...	bias:not printed in abstract | dT:TL = 321 K confirmed in abstract for propo...	confirmed
Thermal-Aware Shallow Trench Isolation Design Optimization for Minimiz... (IEEE, 2019) DOI: 10.1109/ted.2018.2882577	sub-10nm bulk FinFET、SOI FinFET 與 vertical FET(各含 HP/LP 兩種...	bulk FinFET:全 SiO2 STI 基準之 ION degradation rate = 7.8%;改用全 Al2O3 降至 3%(HP 方案,IOFF +36%);上層 20nm SiO2 / 下層 80nm Al2O3 混合降...	bias:未於已讀文本載明(推測為節點標稱 VDD,未證實) | dT:TL,max 隨 STI 設計變化(數值在全文,未讀取)	confirmed
Study of Self-Heating Effects in Silicon Nano-Sheet Transistors (IEEE, 2018) DOI: 10.1109/edssc.2018.8487097	Si lateral GAA stacked nanosheet FET (single & multi-chann...	FinFET: 2.4% ON-current degradation from SHE; NSFET: 1.8% (adjacent, label separately) — both CONFIRMED verbatim from ab...	bias:not stated in abstract (DC TCAD on-state p... | dT:not in abstract; full text not accessed	confirmed
3D Coupled Electro-Thermal FinFET Simulations Including the Fin Shape ... (IEEE, 2014) DOI: 10.1109/sispad.2014.6931615	奈米級矽 SOI FinFET(GARAND 模擬,含 bulk FinFET 範例;非矩形真實 fin 形狀)	圖層級;% 數字未印於可取得文字(Enlighten 無全文、IEEE 阻擋)——引用數字前必讀原文 ID-VD 圖	bias:not accessed | dT:本文數值未取得;同系列 SUPERTHERM E/TED 2015 姊妹作:κ 降低後 ...	confirmed
Simulation of self-heating effects in 30nm gate length FinFET (IEEE, 2008) DOI: 10.1109/ulis.2008.4527143	30nm gate length SOI FinFET	未取得——全文付費牆內；候選條目未宣稱任何 %，摘要亦無數字	bias:未取得（付費牆）| dT:未取得（付費牆）	confirmed
Role of mechanical stress on the electrothermal and OFF state current ... (非, 2026) DOI: 10.1038/s41598-026-46949-1	Silicon bulk FinFET, 3 nm node, Lg=16 nm, fin height 64 nm...	n-FinFET: ID reduced 10.6% with SHE (0% carbon S/D), 8.2% with 10% carbon; p-FinFET: ID reduced 21.6% with SHE at 90% Ge...	bias:VG = VD = 0.7 V (saturation), TA = 300 K | dT:Peak lattice T: n-FinFET 363 K at 10% carb...	confirmed
A modeling method for self-heating effect of FinFET with wide applicat... (非, 2025) DOI: 10.1088/1402-4896/adb2f1	14 nm FinFET(n 與 p),模型延伸驗證至 10 nm 與 7 nm 製程	IDS 最大衰減:pFinFET 8.91%、nFinFET 7.26%(含 SHE vs 無 SHE)@VDS=0.7 V(飽和區)	bias:VDS=0.7 V(兩個 % 對應之偏壓,已從 IOP 全文頁確認) ;VGS 掃描 ... | dT:最高 lattice temperature 399.9 K(ΔT~100 K @T...	confirmed
Self-heating effect in nanoscale SOI Junctionless FinFET with differen... (非, 2024 (arXiv v1); original conference paper 2021) DOI: https://arxiv.org/abs/2403.06271	SOI junctionless FinFET; Lg 10-40 nm, Si channel thickness...	Figure-level only, no printed %. Verbatim quote confirmed in full text: 'SHE in the nanoscale SOI JL FinFET induces subs...	bias:VDS = 0.75 V for Fig. 3 ID-VG comparison (... | dT:Lattice T increases with Lg (Fig. 5) and w...	confirmed
Impact of Variation in Fin Thickness and Self-Heating on the Output Ch... (非, 2024 (online 2024-01-04; print issue April 2024)) DOI: 10.1007/s12633-023-02835-3	Triangular-gate silicon FinFET (TG-FinFET), fin thickness ...	% still not obtained — abstract numerals are truncated in BOTH Springer render and colab.ws mirror ('SHE causes an avera...	bias:not extractable from accessible sources | dT:not extractable from accessible sources	plausible_unveri
Physical insights of interface traps and self-heating effect on electr... (非, 2024) DOI: 10.1088/1402-4896/ad16b0	SOI Dual-Material-Gate (DMG) FinFET, Lg = 16 nm(convention...	SHE 單獨:25.03% 汲極電流退化(conventional);SHE+traps 合併: 30.4%/40.96%(conv, uniform/Gaussian)、45.2%/54.5%(overlap)、37.2%/52.8%(un...	bias:VDS = 0.5 V, VGS 掃 0->1.5 V(IOP 頁面核實)。注意: VG... | dT:摘要無數值(已核實)	confirmed
Analysis of thermal stability in underlap and overlap DMG FinFETs incl... (非, 2024) DOI: 10.1016/j.mejo.2024.106152	SOI DMG FinFET(gate underlap / overlap 通道工程;與 ad16b0 同組、Lg...	SHE 造成之最大 ON-current 下降 41.73%(gate-overlap 設計,但 overlap 之絕對 ION 仍最大);最大 off-current 下降 69.01%(underlap) ——兩數值經出版者摘要文字核實	bias:摘要未載(全文未讀取) | dT:摘要無數值	confirmed
Impact of self-heating on RF/analog and linearity parameters of DMG Fi... (非, 2023) DOI: 10.1016/j.mejo.2023.105765	SOI DMG FinFET(conventional / overlap / underlap)	摘要無 Ion %(已核實);gm,max 退化 8% / 28.3% / 6.9%,gd 退化 30.3% / 40.9% / 33.3%(conventional / overlap / underlap)——經摘要原文核實	bias:摘要未載(全文未讀取) | dT:摘要無數值	confirmed
An Investigation into the Comprehensive Impact of Self-Heating and Hot... (非, 2022) DOI: 10.3390/electronics11172753	22 nm n-channel bulk FinFET, L=22 nm, fin height 40 nm, fi...	Figure-level only, no % printed: 'the drain current in the linear region is almost unchanged, while the drain saturation...	bias:Vds 0-1 V, Vgs 0-1 V (Table 2), TA = 300 K | dT:Max lattice T rises 314.37 K -> 360.76 K a...	confirmed

論文 / IEEE / 年 / DOI	元件	飽和 Ion 下降 % / 關鍵數字	偏壓 / ΔT	查證
Dynamic Self-Heating Effects of Bulk and SOI FinFET with Realistic Dev... (非, 2015) DOI: 10.5573/ieie.2015.52.10.064	Bulk FinFET 與 SOI FinFET(realistic 工業級 3D 結構)	figure-level;已證實摘要僅定性:穩態下 SOI FinFET 驅動電流退化較 bulk 嚴重;高速邏輯操作下 SOI 之動態 SHE 與 bulk 相當(realistic 結構尤然),無任何 %	bias:摘要未載 | dT:摘要無數值	confirmed

三、電熱 vs 等溫 — 相鄰 nanosheet / GAA（14 篇，趨勢對照用）

論文 / IEEE / 年 / DOI	元件	飽和 Ion 下降 % / 關鍵數字	偏壓 / ΔT	查證
Influences of Source/Drain Extension Region on Thermal Behavior of Sta... (IEEE, 2024) DOI: 10.1109/ted.2024.3351596	Stacked nanosheet FET with S/D extension length LEXT swept...	~10% ION degradation attributed to SHE at LEXT = 8 nm (longer extension lowers absolute ION but carries a smaller SHE pe...	bias:not stated in abstract | dT:not printed in abstract	confirmed
Electro-Thermal Characteristics of Junctionless Nanowire Gate-All-Arou... (IEEE, 2023) DOI: 10.1109/ted.2023.3268249	sub-5 nm 節點 junctionless 奈米線 GAA(SOI)	SHE 造成約 10% ON-current 劣化——摘要逐字確認:『SHE rises the peak of TL, causing ~10% ON-current [degradation]』	bias:on-state(摘要未列 VGS/VDS 數值) | dT:通道內非均勻 TL 峰值(摘要無數值)	confirmed
Device Design Aware and Interface Thermal Resistance Assisted Self-Hea... (IEEE, 2022) DOI: 10.1109/icee56203.2022.10117683	GAA 奈米片 FET(NSHFET),並與 SOI FinFET、bulk FinFET 做電熱特性公平比較	figure-level(摘要逐字證實以 drain current 百分比變化比較 SOI FinFET / bulk FinFET / NSHFET,數值僅在內文圖表,未能取得)	bias:未載於摘要(未讀取全文) | dT:空間晶格溫度梯度(數值在全文,未讀取)	confirmed
Geometrical influence on Self Heating in Nanowire and Nanosheet FETs u... (IEEE, 2020) DOI: 10.1109/edtm47692.2020.9117971	Stacked nanowire vs nanosheet GAA-FETs (Sentaurus TCAD), s...	SHE on-current degradation 3.01% (stacked nanowires) vs 6.42% (single nanosheet) at same output current; optimized stack...	bias:not stated in abstract | dT:not printed in abstract	confirmed
Self-Heating and Electrothermal Properties of Advanced Sub-5-nm Node N... (IEEE, 2020) DOI: 10.1109/led.2020.2998460	Sub-5-nm 節點 GAA nanoplate(nanosheet)FET;變化堆疊通道數、金屬閘極厚度、通道寬...	圖級,% 未印於摘要;on-current degradation 隨堆疊數/閘金屬厚度/寬度分析,但退化的比較基準(ET vs isothermal 或幾何參照)無法自摘要確認	bias:摘要未載明 | dT:圖級 (TL,max 有分析),未取得	plausible_unveri
Analysis of Self Heating Effect in DC/AC Mode in Multi-Channel GAA-Fie... (IEEE, 2019) DOI: 10.1109/ted.2019.2942074	三通道堆疊奈米線 GAA FET(SNU Shin 團隊)	DC 模式 Ion 劣化 3.8%(於 ΔTmax=65 K)——摘要逐字確認:『as ΔTmax increases to 65 K, the transistor suffers from an Ion degradation of 3...	bias:DC on-state(摘要未列 VGS/VDS 數值);AC 依頻率(飽和於約 4... | dT:DC ΔTmax=65 K;AC 4 GHz 時 ΔTmax~40 K;duty 2...	confirmed
Analysis on Self-Heating Effects in Three-Stacked Nanoplate FET (IEEE, 2018) DOI: 10.1109/ted.2018.2862918	Three-stacked nanoplate (nanosheet) FET; TCAD calibrated t...	figure-level only; no percentage printed in abstract	bias:not stated in abstract | dT:not printed in abstract	confirmed
The Impact of Self-Heating on Single-Event Transient Effect in Triple-... (非, 2026 issue (Electronics 15(1), January 2026); Crossref records online year 2025) DOI: 10.3390/electronics15010085	Triple-layer stacked Si nanosheet GAA FET, Lg = 12 nm, VDD...	Ion suppressed by ~1.81% with SHE; on/off ratio -1.82% — CONFIRMED verbatim via Google-indexed full text: 'the self-heat...	bias:VDD = 0.65 V, VGS swept 0-0.65 V, TA = 300... | dT:~472 K lattice temperature corroborated: s...	confirmed
Systematic performance benchmarking of nanosheet and FinFET: An intrin... (非, 2025) DOI: 10.1016/j.microrel.2025.115588	Multi-fin silicon FinFET vs vertically stacked nanosheet F...	Not obtained — no % accessible (abstract-level only; ScienceDirect 403 blocks full text)	bias:not extracted (full text not accessed) | dT:not extracted	plausible_unveri
Accurate Evaluation of Electro-Thermal Performance in Silicon Nanoshee... (非, 2024) DOI: 10.3390/nano14121006	sub-3 nm 節點 Si 堆疊奈米片 FET(n 與 p),Lg=12 nm、WNS=25 nm、TNS=5 n...	圖級、無單一 %:Ion_SHE 在 Vco 以上低於 Ion_noSHE(Vco=+0.8 V nFET / -0.65 V pFET,全文逐字確認);Vco 以下 SHE 略增電流。方案間差:TIS 之 Ion_SHE 比 BOX 高 ...	bias:VDD=0.7 V(Table 1),熱邊界 heat sink 300 K;DC ... | dT:DC:TIS 比 BOX 低 59.6 K(n)/50.4 K(p);RO 暫態:T...	confirmed
Trap and self-heating effect based reliability analysis to reveal earl... (非, 2023) DOI: 10.1016/j.sse.2022.108546	Vertically three-stacked Si nanosheet FET, LG = 12 nm, she...	NO numeric % anywhere in full text. Fig. 2(c) (p.3) shows ION degradation in IDS-VGS with vs without SHE (baseline) at V...	bias:VDS = 0.7 V throughout; VGS swept 0-0.7 V;... | dT:Not printed numerically; hotspot located a...	confirmed
The Impact of Ambient Temperature on Electrothermal Characteristics in... (非, 2023) DOI: 10.3390/nano13222971	3 層垂直堆疊奈米片電晶體、1-32 個橫向 stack;模擬單元 Lg=16 nm、WNS=20 nm、TNS=6...	圖級、無印出 %:Fig. 10b 顯示高 VGS 時含 SHE 之 IDS 低於不含 SHE;全文確認無任何明確 SHE 電流下降 %(僅正規化 ΔIDS/IDS 形式)	bias:VGS=VDS=0.7 V、TA=300 K(SHE 數值);另有 TA 掃描 30... | dT:ΔTmax=149 K(Nstack=1、VGS=VDS=0.7 V、Tamb=30...	confirmed
Investigation of Analog/RF and linearity performance with self-heating... (非, 2023) DOI: 10.1016/j.mejo.2023.105904	垂直堆疊 GAA nanosheet FET(3 片);ADJACENT,非 FinFET	純 SHE(有/無 lattice heating)之 Ion % 仍為圖形層級,摘要未印出。注意:摘要印有「ON current 降低 9.4%」但該數字對應 250->400 K 溫度變化(含 SHE 之累積效應),依分類法應歸 ta_s...	bias:摘要未載明個別偏壓;溫度範圍 250-400 K | dT:摘要未給出	confirmed
Self-heating effect on logic performance of 6T-SRAM based on CFET devi... (非, 2022) DOI: 10.35848/1347-4065/ac3c1b	CFET (vertically stacked N-over-P): fin-based CFET (FBC-CF...	Device-level Ion % NOT reported — confirmed by reading article meta/abstract text; paper quantifies circuit level only: ...	bias:SRAM read/write operation; RNM -29 mV figu... | dT:~10 K temperature rise under a single read...	confirmed

四、量測（pulsed-vs-DC / TRE）— SHE 的實測定義（8 篇）

實體元件的 SHE 效應 = DC（完全加熱）與 pulsed/fast I-V（抑制加熱）之差。多數 FinFET % 為圖層級；脈寬需 □1 ns 才除得淨 SHE（τ_th~10-20 ns）。

論文 / IEEE / 年 / DOI	元件	飽和 Ion 下降 % / 關鍵數字	偏壓 / ΔT	查證
In-Situ Monitoring of Self-Heating Effect in Aggressively Scaled FinFE... (IEEE, 2020) DOI: 10.1109/irps45951.2020.9129591	Aggressively scaled SOI FinFETs (Zhejiang Univ., Yi Zhao g...	not printed in accessed material; abstract confirms SHE is suppressed when heating time is short and cooling time suffic...	bias:stress-phase bias with heating/cooling int... | dT:3-D channel temperature-rise maps vs heati...	confirmed
Impact of Self-Heating Effect on Transistor Characterization and Relia... (IEEE, 2019) DOI: 10.1109/jeds.2019.2911085	Lg=20nm bulk 與 SOI FinFET、14nm 多鰭 FinFET（in scope）；Lg=30nm...	FinFET：圖層級（Fig. 4，四種速度之 ID-VD，速度愈快 Ion 愈高；SOI FinFET 之 pulsed-vs-DC on-current 差異為所有受測結構中最嚴重），全文未印出單一 FinFET %。文中唯二印出的 %...	bias:HCI 應力 VG=VD=1.1V；彈道傳輸量測 298-423K；I-V 於 0.... | dT:FinFET 通道達熱平衡升溫約 60K（planar bulk 僅約 10K）；通...	confirmed
Ultra fast (<1 ns) electrical characterization of self-heating effect ... (IEEE, 2017) DOI: 10.1109/iedm.2017.8268520	14nm 節點商用 Si bulk FinFET	未取得——IEDM 全文付費牆內，pulsed-vs-DC ID 差異為圖層級；引文脈絡（Semantic Scholar citation contexts）亦查無被引用的 %。姊妹篇 JEDS 2019 顯示 FinFET dT~60K...	bias:多組 VGS/VDS 組合（具體值未取得）；HCI 部分為 GHz 隨機訊號 | dT:摘要層級：任意工作條件下可萃取暫態與穩態通道溫度；具體 K 值在付費牆內（姊妹篇 J...	confirmed
Investigation of Self-Heating Effect on Ballistic Transport Characteri... (IEEE, 2017) DOI: 10.1109/ted.2016.2646907	ultrascaled Si FinFETs（研究元件；Lg 未於可及文本揭露）	未取得——付費牆；摘要僅定性稱 DC 法因嚴重 SHE 使彈道傳輸參數萃取出現 essential discrepancies，% 在圖層級	bias:未取得（付費牆）| dT:未取得（付費牆）	confirmed
A Thermal-Aware Device Design Considerations for Nanoscale SOI and Bul... (IEEE, 2016) DOI: 10.1109/ted.2015.2502062	sub-22 nm SOI 與 bulk Si FinFET	摘要無 %——pulse rise-time 法取得等溫特性(隱含 pulsed vs DC 電流差在圖級);全文未能存取	bias:摘要未載明 | dT:未取得	confirmed
Hot carrier reliability characterization in consideration of self-heat... (IEEE, 2016) DOI: 10.1109/irps.2016.7574505	Samsung production FinFET (14nm-class)	none; no Ion % — reliability-lifetime class metric only	bias:not accessed (full text paywalled) | dT:not accessed	confirmed
A Reflection-Based Ultra-Fast Measurement Method for the Continuous Ch... (非, 2025) DOI: 10.3390/electronics14132634	Commercial-ready scaled short-channel transistors, describ...	Figure-level only: Fig. 4 shows continuous ID degradation vs time (first ~50 ns) at chuck temperatures 223-373 K; no ste...	bias:VG = 700 mV, VD = 690 mV (Fig. 5a heat-fre... | dT:Real-time T(t) extracted via Iheat-free = ...	value_corrected
Experimental investigation of self heating effect (SHE) in multiple-fi... (非, 2014) DOI: 10.1088/0268-1242/29/11/115021	金屬閘多鰭 SOI FinFET：Lg 90nm-5um、Wfin 20-35nm、Hfin 58nm、2-50 平...	n-channel >10%、p-channel >7% 飽和電流劣化（於最短 Lg=90nm 時最嚴重）——IOP 全文確認	bias:飽和區：n-channel VGS=VDS=1.5V；p-channel VGS=V... | dT:各幾何之 Rth/dT 於圖中給出；Rth 隨 Lg 縮短、鰭數減少、鰭寬變窄而增加...	confirmed

五、熱阻 / 溫升導向（Rth、ΔT；45 篇）

主要提供 Rth / ΔT（thermode SurfaceResistance 的校準標的），未必印出電流 %。single-fin Rth 約 1-4 MK/W；多鰭多指 RF 結構約 34 kK/W。

論文 / IEEE / 年 / DOI	元件	Rth / ΔT / 關鍵數字	偏壓 / 其他	查證
Multiscale Thermal Simulation for GAAFET With First-Principles-Based B... (IEEE, 2025) DOI: 10.1109/ted.2025.3592887	堆疊奈米片 GAAFET(約 10 nm 特徵尺寸)	摘要未給數值;僅定性指出 gray BTE/HDE 低估主動區溫度、需 nongray BTE	無(摘要未載)	confirmed
Transient Thermal Model of Advanced Nanotransistors: A Case Study of C... (IEEE, 2025) DOI: 10.1109/ted.2025.3574124	monolithic CFET (stacked nFET-on-pFET) as case study; mode...	no values in abstract; Fourier-vs-DPL transient temperature differences shown figure-level only	not applicable / not accessible	confirmed
Impact of Back End of Line (BEOL) and Ambient Temperature on Self-Heat... (IEEE, 2024) DOI: 10.1109/edtm58488.2024.10511833	twin 奈米線 GAA FET(junction less vs inversion mode)	摘要未載	摘要未列 VGS/VDS;RTH 與 TA 為掃描變數	reclassified
Understanding the Self-Heating Effects Measured With the AC Output Con... (IEEE, 2024) DOI: 10.1109/ted.2024.3469187	advanced-node bulk silicon FinFETs, short- and long-channe...	abstract confirms: AC-gout overtemperature = area-averaged overtemperature across active area; largely underestimates pe...	gate-voltage dependence discussed; specific VGS/VDS values figure-level, not accessed	confirmed
Influence of the Gate Oxide and Back Oxide Material Types on Self-heat... (IEEE, 2024) DOI: 10.1109/edm61683.2024.10615156	SOI junctionless FinFET（Sentaurus 3D TCAD）	晶格溫度隨閘極氧化層厚度增加大致單調下降、隨背氧化層厚度增加單調上升（歸因於材料熱導率差異與接觸面積變化；數值為圖級）	未取得（全文未讀）	confirmed
Theoretical Study of Self-Heating-Induced Thermal Stress Effects on Qu... (IEEE, 2023) DOI: 10.1109/ted.2023.3280865	p-type ultrathin-body silicon FinFET; multiple crystal ori...	not printed in abstract	not accessed	confirmed
Localized thermal effects in Gate-all-around devices (IEEE, 2023) DOI: 10.1109/irps48203.2023.10117903	Intel gate-all-around (RibbonFET-class) devices, thermal r...	temperature-rise data collected at transistor scale; values figure-level, not in abstract	not accessed	confirmed
Self-Heating in iN8-iN2 CMOS Logic Cells: Thermal Impact of Architectu... (IEEE, 2022) DOI: 10.1109/vlsitechnologyandcir46769.2022.9830228	imec 節點標準邏輯 cell:FinFET(iN8-iN5)、nanosheet(iN5/iN3)、forksh...	已自摘要查證:excess channel heating(相對 cell boundary)baseline 0.8-1.6 K;turbo 10-19 K	cell 級操作情境:baseline = 0.7VDD @ 2 GHz clock;turbo = 1.4VDD @ 6 GHz clock	confirmed
Self-Heating and Thermal Network Model for Complementary FET (IEEE, 2022) DOI: 10.1109/ted.2021.3130010	CFET (vertically stacked complementary n/p device); array-...	figure-level; thermal-network lattice T drives lifetime prediction	not accessed	confirmed
Thermal Conductivity Model to Analyze the Thermal Implications in Nano... (IEEE, 2022) DOI: 10.1109/ted.2022.3208848	SOI-based junctionless nanowire (JL-NW) FET at sub-5-nm te...	not in abstract	not accessible	confirmed
FEOL Self-heating and BEOL Joule-heating Effects of FinFET Technology ... (IEEE, 2020) DOI: 10.1109/iirw49815.2020.9312859	Advanced bulk FinFET (taller/narrower fin, shorter CPP) wi...	Abstract-verified: SHE aggravated by compact layout footprint; taller/narrower fin and shorter CPP give higher dT_sh; st...	not accessed	confirmed
Impact of Geometry, Doping, Temperature, and Boundary Conductivity on ... (IEEE, 2020) DOI: 10.1109/tdmr.2020.2964734	14-nm bulk 與 SOI FinFET(實驗 ID-VG 校準之 TCAD)	not accessed(摘要無數值)	not accessed	confirmed
Advanced Self-heating Model and Methodology for Layout Proximity Effec... (IEEE, 2020) DOI: 10.1109/irps45951.2020.9128322	先進 FinFET 技術（Samsung foundry）	ΔTsh 隨佈局鄰近之分佈（摘要以 ΔTsh 為核心量，無具體數值）	摘要未給；全文未讀	confirmed
Impact of Self-heating on Performance and Reliability in FinFET and GA... (IEEE, 2019) DOI: 10.1109/isqed.2019.8697786	7nm bulk/SOI FinFET 與 5nm GAAFET 標準元件/邏輯閘子結構（GAA 部分為 adjac...	邏輯閘平均升溫：7nm SOI FinFET +12K、bulk FinFET +5K、5nm GAAFET +17K（摘要與全文 Fig. 7 一致）	電路級 workload（activity factor 加權），非單一 VGS/VDS 偏壓點	reclassified
Investigation of self-heating effect on stacked nanosheet GAA transist... (IEEE, 2018) DOI: 10.1109/vlsi-tsa.2018.8403821	水平堆疊三層 GAA nanosheet 電晶體(北京大學組),含簡化 BEOL	圖級數據,未能取得	摘要未載明	reclassified
Layout Design Correlated With Self-Heating Effect in Stacked Nanosheet... (IEEE, 2018) DOI: 10.1109/ted.2018.2825498	5 nm 節點水平堆疊三層 GAA nanosheet 電晶體,單元件至元件陣列	圖級,未取得	摘要未載明	confirmed
Modeling of FinFET Self-Heating Effects in multiple FinFET Technology ... (IEEE, 2018) DOI: 10.1109/vlsit.2018.8510657	Production bulk FinFETs across multiple technology generat...	Abstract confirms taller/narrower fins in newer generations increase FSH; per-generation dT values are figure-level only...	not accessed	confirmed
Ambient temperature and layout impact on self-heating characterization... (IEEE, 2018) DOI: 10.1109/irps.2018.8353640	Bulk FinFET (14nm-class), multi-fin/multi-finger layouts	Abstract-verified new number: ambient temperature can affect the self-heating MEASUREMENT by up to 70% (first such demon...	not accessed	value_corrected

論文 / IEEE / 年 / DOI	元件	Rth / ΔT / 關鍵數字	偏壓 / 其他	查證
Bottom-up methodology for predictive simulations of self-heating in ag... (IEEE, 2018) DOI: 10.1109/irps.2018.8353650	aggressively scaled bulk FinFET process technologies (GF 1...	not accessed	not accessed	confirmed
Self-Heating Effects Investigation on Nanoscale FinFET and Its Thermal... (IEEE, 2018) DOI: 10.1109/icsict.2018.8564853	sub-14 nm FinFET（TCAD，以實驗資料校準）	熱散逸對 S/D extension 尺寸敏感；摻雜濃度相關之溫度變化（可藉縮短 extension 降低）；fin 寬/高影響電特性、hotspot 位置與集總 Rth	未取得（全文未讀）	confirmed
A Novel Approach to Localize the Channel Temperature Induced by the Se... (IEEE, 2018) DOI: 10.1109/edtm.2018.8421479	14 nm high-k/metal-gate bulk FinFETs (UMC), n and p	pFinFET channel ~170 K hotter than nFinFET (abstract-confirmed; attributed to embedded SiGe S/D heat dissipation); hotte...	saturation-region operation for RTN thermometry; exact VGS/VDS not in accessed material	confirmed
Analysis of DC Self Heating Effect in Stacked Nanosheet Gate-All-Aroun... (IEEE, 2018) DOI: 10.1109/edtm.2018.8421495	Vertically stacked nanosheet GAA transistor (SNU/KNUT TCAD...	figure-level per abstract ('great lattice temperature variations'); values not in abstract	DC bias (values not in abstract; full text not accessed)	reclassified
Modeling of Effective Thermal Resistance in Sub-14-nm Stacked Nanowire... (IEEE, 2018) DOI: 10.1109/ted.2018.2863730	sub-14nm FinFET 與堆疊奈米線 FET(含大型 layout:多鰭、多 gate finger)	Rth vs 幾何設計變數(鰭數、finger 數、奈米線堆疊數/尺寸);適用於大 layout 之 Rth 估算	不適用(Rth 模型)	confirmed
Self-heating measurement methodologies and their assessment on bulk Fi... (IEEE, 2017) DOI: 10.1109/iirw.2017.8361226	bulk FinFET (GlobalFoundries technology), wafer-level test...	no absolute dT/Rth printed in abstract; key quantified result: self-heating underestimated by 35% when sensed at a neigh...	not verified (full text not accessed)	confirmed
Self-heating in FinFET and GAA-NW using Si, Ge and III/V channels (IEEE, 2016) DOI: 10.1109/iedm.2016.7838425	Si (and Ge, III-V) FinFET and GAA nanowire, industry-relev ...	Figure-level dT/Rth comparisons across architectures and processing options; not accessed	not accessed (closed access)	value_corrected
Self-Heating Measurement of 14-nm FinFET SOI Transistors Using 2-D Tim... (IEEE, 2016) DOI: 10.1109/ted.2016.2537054	14-nm-node SOI FinFET (IBM), silicon; multiple geometries/...	figure-level only; not printed in abstract	not printed in abstract	confirmed
Characterization of self-heating leads to universal scaling of HCI deg... (IEEE, 2016) DOI: 10.1109/irps.2016.7574506	Multi-fin SOI FinFETs (sub-14 nm class), fin-count series	fin-count-dependent lattice temperature rise (values figure-level, not in accessed text)	HCI stress bias; Tsub and AC frequency swept; exact values not in accessed material	confirmed
Self-heating on bulk FinFET from 14nm down to 7nm node (IEEE, 2015) DOI: 10.1109/iedm.2015.7409678	Bulk Si (and strained-Ge) FinFET, 14nm/10nm/7nm nodes	Heat confinement increases ~20% per 0.7x node scaling for Si channel, and an additional ~57% for strained-Ge channel (ab...	not accessed (closed access)	confirmed
Investigation of Self-Heating Effect on Hot Carrier Degradation in Mul... (IEEE, 2015) DOI: 10.1109/led.2015.2487045	Multiple-fin SOI FinFETs (fin-number split)	higher T rise at larger fin number (numeric values figure-level, not in accessed text)	HCD stress bias; exact values not in accessed material	confirmed
Thermal behavior of self-heating effect in FinFET devices acting on ba... (IEEE, 2015) DOI: 10.1109/irps.2015.7112696	先進 bulk FinFET 技術（TSMC），含多金屬層金屬電阻溫度感測器測試結構	摘要無數值；內容涵蓋隨供電元件數的溫度飽和、各金屬層溫升、SHE 與 Joule heating 耦合效應	摘要未給；全文未讀	confirmed
Self-heating effect in FinFETs and its impact on devices reliability c... (IEEE, 2014) DOI: 10.1109/irps.2014.6860642	Production FinFET (16nm-class, bulk) vs planar reference	Not printed in abstract; dT numbers remain full-text-only — do not quote until read	Abstract recommends HCI stress condition Vg = 0.6~0.8Vd to keep power/temperature reasonable (abstra...	confirmed
Experimental validation of self-heating simulations and projections fo... (IEEE, 2014) DOI: 10.1109/irps.2014.6861186	實測為平面（FD）元件；FinFET 部分僅為有限元素模擬展望	摘要無數值（未提 Rth/dT 具體數字）	摘要未給；全文未讀	confirmed
Analytical Thermal Model for Self-Heating in Advanced FinFET Devices W... (IEEE, 2013) DOI: 10.1109/tcad.2013.2248194	SOI/bulk multifin FinFET structures (analytical model vali...	Rth/dT sensitivity vs fin geometry is figure-level; not accessible without full text	not applicable (thermal model paper)	confirmed
Self-heat reliability considerations on Intel's 22nm Tri-Gate technolo... (IEEE, 2013) DOI: 10.1109/irps.2013.6532036	Intel 22nm bulk tri-gate (FinFET) production technology	not accessed (full text paywalled)	not accessed	confirmed
Experimental analysis and modeling of self heating effect in dielectri... (IEEE, 2013) DOI: https://ieeexplore.ieee.org/document/6576663	dielectric-isolated (SOI) FinFET vs 45nm planar SOI	not accessed; abstract: scaled FinFET-on-dielectric shows higher normalized thermal resistance than 45nm planar SOI, as ...	not accessed	confirmed
Assessment of NBTI in Presence of Self-Heating in High-k SOI FinFETs (IEEE, 2012) DOI: 10.1109/led.2012.2213572	high-k SOI FinFET (silicon)	not quantified in abstract	VERIFIED from abstract: no-SH NBTI stress at Vgs=-2V, Vds=0V at T=25C and 125C; with-SH stress at ro...	confirmed
RF Extraction of Self-Heating Effects in FinFETs (IEEE, 2011) DOI: 10.1109/ted.2011.2162333	n-channel SOI FinFETs, various fin width / fin number / fi...	figure-level, not verified	not verified (full text not accessed)	confirmed

論文 / IEEE / 年 / DOI	元件	Rth / ΔT / 關鍵數字	偏壓 / 其他	查證
Thermal-aware device design of nanoscale bulk/SOI FinFETs: Suppression... (IEEE, 2011) DOI: 10.1109/iedm.2011.6131672	nanoscale bulk FinFET vs SOI FinFET (electrothermal simula...	not quantified in accessible text; abstract: bulk FinFET lattice temperature significantly lower than SOI FinFET owing t...	not accessed	confirmed
Modulating self-heating effects in FinFETs through doping engineering (非, 2026) DOI: 10.1063/5.0309190	先進節點 Si FinFET	Thomson heat 貢獻最大元件溫升約 16%;降低延伸區摻雜梯度後峰值 Thomson heat 減 66%、熱點溫升降 13%——三個數字皆與摘要逐字核實相符	saturation bias vs 典型 CMOS 操作偏壓(摘要未給具體電壓值)	reclassified
Effect of Gate Oxide and Back Oxide Materials on Self-Heating Effect i... (非, 2025) DOI: 10.26565/2312-4334-2025-3-35	n 通道矽 FinFET(SOI),TiN 閘極, Lgate=10 nm,通道厚 Tsi=9 nm,通道寬 Wb=2...	通道中心晶格溫度:Fig.3 約 320-380 K(各 gate/back oxide 組合,tox 1.0-1.5 nm);Fig.5 隨 TBOX 10->1000 nm 由約 300 K 升至約 550 K(HfO2 BOX 最高、S...	Fig.5(BOX 掃描):VDS=0.75 V、VG=1.5 V,gate oxide HfO2 tEOT=1.2 nm	value_corrected
Comparative study of the self-heating effect in the accumulation and i... (非, 2024) DOI: https://arxiv.org/abs/2402.10858	SOI FinFETs, Lg = 10 nm, Tsi = 9 nm, fin width Wb = 22 nm,...	Confirmed from Fig. 4: channel lattice T ~368.5-370.5 K (inversion FinFET) vs ~455-458 K (JL FinFET) along channel at Vd...	Fig. 4 temperature readout at Vd = 0.75 V, Vg = 1.6 V (printed in Fig. 4 caption — improves on candi...	confirmed
Integrated modeling of Self-heating of confined geometry (FinFET, NWFE... (非, 2018) DOI: 10.1016/j.microrel.2017.12.034	Sub-20 nm silicon FinFET, nanowire FET, nanosheet FET (con...	not extracted	n/a (full text not accessed)	confirmed
Impact of self-heating effect on the performance of hybrid FinFET (非, 2018) DOI: 10.1016/j.mejo.2018.04.015	Hybrid FinFET (silicon), variable channel length, fin widt...	Lattice T 726.5 -> 495.6 K vs pitch (up to ~430 K above ambient at tight pitch)	not stated at abstract level	confirmed
Analysis on Self-Heating Effect in 7 nm Node Bulk FinFET Device (非, 2016) DOI: 10.5573/jsts.2016.16.2.204	7nm node 非矩形 bulk FinFET(ITRS 2013 LP),含 W contact via 與 C...	ΔTLmax~37.1 K(hotspot TL=337.1 K @TA=300K);兩元件互連(一開一關)時 hotspot 329.3 K;兩管串聯時 320.4 K(Tr.1 實際壓降僅約 0.59 V,非等偏壓比較)	VGS=VDS=0.78 V, TA=300 K(Figs.5-7 標注);DIBL=37.0 mV/V	confirmed
DC self-heating effects modelling in SOI and bulk FinFETs (非, 2015) DOI: 10.1016/j.mejo.2015.02.003	奈米級 SOI 與 bulk Si FinFET, 單 fin 至 multi-fin/multi-finger	not accessed	not accessed	confirmed

六、可靠度 / 其他指標（HCI、BTI、TDDB、gm/gd、電路級；34 篇）

自熱對可靠度壽命與類比/RF 指標的影響，含 gm/gd 退化。

論文 / IEEE / 年 / DOI	元件	關鍵數字 / 指標	偏壓 / ΔT	查證
Impact of Electro-Thermal Transport on HCI and BTI Lifetime of Twin Na... (IEEE, 2025) DOI: 10.1109/tdmr.2025.3570616	twin Si GAA nanowire FETs compared in two operational mode...	junctionless shows ~15.2% better thermal reliability and ~26.7%/37.6% better HCI/BTI lifetime than inversion-mode (DC); ...	bias:not accessible (full text paywalled); tran... | dT:not in abstract	confirmed
ML-Driven Transistor Self-Heating Analysis From TCAD to Large IP Circu... (IEEE, 2025) DOI: 10.1109/access.2025.3576161	先進 CMOS 電晶體(TCAD 訓練 ML;搜尋證據顯示方法為 technology-agnostic,涵蓋 FD...	SHE-aware aging guard band:Vth-shift 估計自保守常數 50 mV 精化至 31 mV(38% 縮減為推導值,非原文引句)	bias:not accessed | dT:ML 預測每電晶體 SHE 溫度,對 TCAD 參考之平均誤差 0.1 K	confirmed
On the Output Conductance Dispersion due to Traps and Self-Heating in ... (IEEE, 2024) DOI: 10.1109/esserc62670.2024.10719502	大面積 nMOS：bulk、FDSOI、bulk FinFET 三種架構並列	gDD 頻譜在低/高 VGS 呈相反趨勢；自熱與缺陷（trap）貢獻分離；改善由 gDD 頻散萃取 Rth 的準確度；ZTC 相關異常觀察	bias:低/高 VGS 對比；在 DC 轉移特性溫度靈敏度變號（ZTC）附近的 VGS 觀察...	confirmed
Multiphysics Simulation of Self-Heating-Induced Thermal Stress Effects... (IEEE, 2024) DOI: 10.1109/ted.2024.3430255	p-type GAA stacked nanosheet FET	thermal stress alters band structure, DOS, hole effective mass, hole density, valence subband profiles	bias:not accessible (full text paywalled) | dT:not in abstract	confirmed
Electrothermal Effects on Hot Carrier Injection Reliability of n-Type ... (IEEE, 2024) DOI: 10.1109/ted.2023.3346840	n 型 FinFET(FreePDK15 15nm 製程,九級環形振盪器,含經驗證之 layout)	HCI 引致 TVS(threshold voltage shift)vs 供應電壓與熱環境(thermal surrounding)	bias:RO 振盪電壓(AC 應力,時變),VDD 掃描;細節在全文 | dT:瞬態空間溫度響應(數值在全文,未讀取)	confirmed
Impact of Device-to-Device Thermal Interference Due to Self-Heating on... (IEEE, 2024) DOI: 10.1109/access.2024.3366347	7-nm 節點堆疊奈米片 FET 對(He-FET 加熱源 + 鄰近 Co-FET),矽,VIT Vellore	Co-FET 下層通道電子遷移率較上層低 4.25%;最佳化後 fT = 131.5 GHz、fmax = 435 GHz;最佳 IDS = 30 nm、STI 深度 = 50 nm	bias:not accessed | dT:摘要未印出 K 數值;全文 PDF 被阻擋未能取得	confirmed
Demonstration of a Nanosheet FET With High Thermal Conductivity Materi... (IEEE, 2023) DOI: 10.1109/ted.2023.3241884	GAA stacked nanosheet FET with crystalline diamond-like-ca...	geometry dependence (Lg, Tw, TBOX, NS) on thermal and electrical reliability	bias:not accessible (full text paywalled) | dT:lattice-temperature reduction via DLC BOX ...	confirmed
Transient Self-Heating Effects on Mixed-Mode Hot Carrier and Bias Temp... (IEEE, 2023) DOI: 10.1109/ted.2023.3312053	奈米級 p 型 FinFET(實測元件;北大 Runsheng Wang/Ru Huang 團隊)	mixed-mode HCD-BTI 老化模型、電路老化(aging)預測精度	bias:mixed VG/VD 應力條件(細節在全文,未讀取) | dT:瞬態 SHE 溫升(數值在全文,未讀取)	confirmed
Self-heating effect of GAAFET and FinFET for over 2-V applications usi... (IEEE, 2022) DOI: 10.1109/iceic54506.2022.9748640	類比用 FinFET 與 GAAFET（>2 V 操作）TCAD 結構	FinFET vs GAAFET 的 SHE 對比；dummy pattern 熱隔離設計；關鍵字含 thermal noise、CIS/ISP 應用	bias:>2 V 高壓類比應用；確切 VGS/VDS 未取得	confirmed
A Method to Isolate Intrinsic HCD and NBTI Contributions Under Self He... (IEEE, 2022) DOI: 10.1109/ted.2022.3172055	GAA stacked nanosheet (GAA-SNS) p-channel FETs with varyin...	pure NBTI vs pure HCD kinetics separated under SH; end-of-life degradation projected across VG/VD map — supports the 'SH...	bias:multiple VG and VD stress conditions and m... | dT:not in abstract	confirmed
Analysis of Self-Heating Effects in Multi-Nanosheet FET Considering Bo... (IEEE, 2022) DOI: 10.1109/ted.2022.3141327	3-nm-node multi-nanosheet FET (mNS-FET); bulk substrate, P...	SHE-induced lifetime reduction: 15.1-22.1% (logic RO), 53.5-89.9% (two analog circuits, with possible circuit malfunctio...	bias:dynamic circuit operation; VDD not stated ... | dT:channel T rise larger with BO than without...	confirmed
Understanding and Modeling Opposite Impacts of Self-Heating on Hot-Car... (IEEE, 2022) DOI: 10.1109/irps48227.2022.9764515	n 通道 Si FinFET + p 通道奈米線（NW）FET（imec）	摘要逐字證實核心結論：nFinFET 中 SH 對載子傳輸的影響與鍵斷裂熱分量增強互相抵消->HCD 受 SH『輕微抑制』；pNWFET 中兩因素同向->HCD 被 SH 加速	bias:多組 HC 應力條件；具體數值需讀全文	confirmed
Electro-Thermal Performance Boosting in Stacked Si Gate-All-Around Nan... (IEEE, 2021) DOI: 10.1109/ted.2021.3095038	垂直堆疊 Si GAA nanosheet FET,S/D 接觸工程:M0-wrap around S/D epi、...	Rth,eff 與 TL,max 降低 >11%(M0 接觸工程 vs 傳統 M0-epi);nanosheet 層間結溫差降低->inter-sheet VT 差異縮小;M0-trench 為電/熱最佳方案	bias:摘要未載明 | dT:TL,max 降低(隨 Rth,eff 一併 >11% 改善);絕對值在圖級,未取得	reclassified
Self-heating Effects Measured in Fully Packaged FinFET Devices (IEEE, 2021) DOI: 10.1109/miel52794.2021.9569208	已封裝之商用 FinFET 元件	自熱的頻率相依性（動態 SHE）；封裝內測試方法；元件級與系統級溫升相關性	bias:切換操作（頻率相依）；確切偏壓未取得（全文未讀）| dT:電流引致自熱溫升實測至 1 GHz、外推至 3 GHz；全晶片動態功耗溫升亦有評估（...	confirmed
Minimizing Excess Timing Guard Banding Under Transistor Self-Heating T... (IEEE, 2021) DOI: 10.1109/access.2021.3057900	FinFET 技術(先進節點,矽;裝置到電路/處理器層級,KIT)	ZTC 偏壓可達近零 SHE timing guard band;存在 thermal guard band 與降壓性能損失之 trade-off(量化值在全文圖表層級,未讀取)	bias:not accessed | dT:摘要未印出;全文量化數字(節點、dT、guard band %)未能取得(scisp...	confirmed
Reliability Modeling and Analysis of Hot-Carrier Degradation in Multip... (IEEE, 2019) DOI: 10.1109/ted.2019.2905053	14 nm SOI n-channel FinFETs (IBM technology), multiple-fin...	SH-enhanced HCD; AC-vs-DC stress frequency dependence; bulk vs interface trap partitioning; empirical SH-aware reliabili...	bias:DC and high-frequency AC HCI stress bias; ...	confirmed
Self-Heating Effects on Hot Carrier Degradation and Its Impact on Logi... (IEEE, 2019) DOI: 10.1109/tdmr.2019.2916230	Bulk FinFETs (GlobalFoundries 14nm-class) + logic ring osc...	Abstract finding: HC degradation modulation due to SH is only significant for logic PFETs at highly accelerated DC condi...	bias:HCI stress, highly accelerated DC vs moder...	confirmed

論文 / IEEE / 年 / DOI	元件	關鍵數字 / 指標	偏壓 / ΔT	查證
New Insight Into Negative Bias Temperature Instability Degradation Dur... (IEEE, 2019) DOI: 10.1109/led.2019.2930077	14-nm bulk p-FinFET (Samsung/SNU, silicon)	NBTI+SHE 下之 Vth 漂移;長時間(10-10^3 s)冪次律時間指數 n 隨|VDS| 增大而下降,證明 DC 應力下 Vth 漂移由 NBTI(而非 HCD)主導	bias:VGS = -1.3 V,VDS 掃至 -1.3 V,應力時間 10 s ~ 10^... | dT:摘要未印出數值	confirmed
An Unique Methodology to Estimate The Thermal Time Constant and Dynami... (IEEE, 2018) DOI: 10.1109/iedm.2018.8614479	advanced TSMC bulk FinFET technology (node unspecified)	thermal time constant of advanced FinFETs; dynamic (AC/duty-cycle) vs static SHE impact on reliability lifetime evaluati...	bias:not verified (full text not accessed) | dT:figure-level, not verified	confirmed
An Experimental Approach to Characterizing the Channel Local Temperatu... (IEEE, 2018) DOI: 10.1109/jeds.2018.2859276	14 nm HK/MG bulk FinFET (n- and p-type), channel lengths s...	Rth increases rapidly as channel length scales to 20 nm (abstract-level)	bias:Saturation Id-Vds; exact VGS/VDS not verif... | dT:Highest temperature at drain edge; pFinFET...	confirmed
Self-Heating Effects on Hot Carrier Degradation and its Impact on Ring... (IEEE, 2018) DOI: 10.1109/iirw.2018.8727093	Bulk FinFETs (GlobalFoundries 14nm-class) + ring oscillato...	Abstract finding: HC degradation modulation due to SH only significant for logic PFETs at highly accelerated conditions;...	bias:HCI stress / RO operation; exact V not acc...	confirmed
Self-heating-aware CMOS reliability characterization using degradation... (IEEE, 2018) DOI: 10.1109/irps.2018.8353541	on-chip FET 陣列（imec；摘要僅稱 FET/VLSI devices，未明寫 FinFET）	摘要明文：『by experimental assessment of thermal resistance and degradation activation energies, we can project a self-heatin...	bias:整個可量測 {Vg,Vd} 偏壓空間（摘要明文）	confirmed
Relaxation of Self-Heating-Effect for Stacked-Nanowire FET and p/n-Sta... (IEEE, 2018) DOI: 10.1109/jeds.2018.2882406	垂直堆疊矽奈米線 FET;p/n-stacked nanowire 6T-SRAM(Tokyo Tech)	p/n-stacked 6T-SRAM 版圖面積約 -15%(vs 傳統 cell)	bias:not accessed(全文未能取得) | dT:摘要僅定性描述奈米線區晶格溫度「considerably decreased」;K ...	confirmed
Investigation of Electrothermal Behaviors of 5-nm Bulk FinFET (IEEE, 2017) DOI: 10.1109/ted.2017.2766214	5 nm node bulk FinFET (scaled per ITRS)	AC delay and BTI figures of merit vs device parameters; S/D contact scheme design improves both performance and reliabil...	bias:not accessed (closed access) | dT:not accessed	confirmed
Electrothermal Effects on Hot Carrier Injection in n-Type SOI FinFET U... (IEEE, 2017) DOI: 10.1109/ted.2017.2728083	50-nm 節點 n 型 SOI FinFET(反相器中之 nFinFET;含 gradual-width fin ...	HCI 引致 Vth shift(PRBS 略大於同頻 AC);gradual-width fin 降低靜態溫度但對動態(AC/PRBS)改善有限	bias:反相器 circuit-speed 偏壓: step/AC/PRBS 訊號,頻率掃描(... | dT:瞬態溫度峰值:PRBS < AC(同頻,因邏輯轉換較少);頻率越高 SHE 越嚴重(...	confirmed
Self-heating impact on TDDB in bulk FinFET devices: Uniform vs Non-uni... (IEEE, 2016) DOI: 10.1109/iirw.2016.7904898	bulk FinFET（GLOBALFOUNDRIES 先進節點）	TDDB 壽命隨 VD 增加而下降，與升溫應力一致；閘壓冪次律在不同 VD 下保持；正常操作下 SHE 有限、電路級 TDDB 未劣化	bias:TDDB 應力於多個 VD 值；確切電壓需讀全文	confirmed
Self-heating and its implications on hot carrier reliability evaluatio... (IEEE, 2015) DOI: 10.1109/irps.2015.7112726	Bulk and SOI FinFET-era technologies (IBM); geometry not s...	SH effect on activation energy Ea; SH-induced HCD enhancement; SH-corrected lifetime methodology; DC-vs-AC (ring oscilla...	bias:HCI stress conditions; exact values not in...	confirmed
Self-heating characterization of FinFET SOI devices using 2D time reso... (IEEE, 2015) DOI: 10.1109/irps.2015.7112672	SOI FinFET（IBM 14nm 世代；個別元件與密集陣列）	off-state 漏電流受自熱調變（定量）；摘要明文宣稱『first direct measurement of the self-heating from individual FinFET devices』	bias:摘要未給；全文未讀 | dT:摘要無數值	confirmed
Impact of source (drain) doping profiles and channel doping level on s... (非, 2025) DOI: 10.1016/j.micrna.2024.208015	SOI FinFET (silicon), constant vs analytical S/D doping pr...	Vth and Ion/Ioff vs S/D doping profile and channel doping level	bias:not extracted (full text not accessed; CC-... | dT:Highest channel-center lattice T for const...	confirmed
A simulation study on optimizing the self-heating and hot carrier effe... (非, 2025) DOI: 10.1007/s12043-025-02998-1	SOI 型矽 FinFET,提出 heterodielectric buried-oxide (HDB) 結構:BO...	熱阻降低、impact ionization 率下降(熱載子抑制)、band-gap narrowing 論證,皆為定性	bias:未於可讀取全文段落中明示 | dT:未給出絕對值(僅圖形層級)	confirmed
Self-Heating Effect Coupled Compact Model to Predict Hot Carrier Injec... (非, 2025) DOI: 10.3390/app15052351	奈米級 bulk FinFET(compact-model 層級;含寬度相依與基板電壓相依)	含 SHE 貢獻之 HCI 退化/元件壽命預測(次要指標:reliability lifetime)	bias:涵蓋多種 stress 條件(不同載子能量區間) | dT:摘要未給出	confirmed
A new characterization model of FinFET self-heating effect based on Fi... (非, 2024) DOI: 10.1016/j.mee.2024.112155	FinFET (n/p); node not stated in abstract-level sources ac...	delta-gm vs delta-T relationship for SHE characterization (gm is a requested secondary metric); circuit-level applicabil...	bias:CORRECTED: abstract states characterizatio... | dT:Characterization accuracy delta-T differen...	value_corrected
Substrate BOX engineering to mitigate the self-heating induced degrada... (非, 2022) DOI: 10.1016/j.mejo.2022.105590	堆疊 GAA nanosheet FET,S/D 下方以 crystalline diamond BOX(BOXNS...	有效熱阻 Reff 降低、ION/IOFF 改善、300-370 K 環溫下晶格溫度波動較小(皆 BOXNS vs SiONS)	bias:摘要未載明;環境溫度變異範圍 300-370 K | dT:diamond BOX 較 SiO2 BOX 晶格溫度降低 13.5 K(摘要數字,...	confirmed
Influence of Joule effect on thermal response of nano FinFET transisto... (非, 2021) DOI: 10.1016/j.spmi.2021.106980	20 nm 矽 FinFET	準彈道熱傳模型精度(優於 DPL 與 C-V 模型)	bias:摘要未載明 | dT:暫態溫度演化為圖形層級,摘要無絕對 dT	value_corrected

七、環境溫度掃描（ta_sweep）— 注意：這不是 SHE 效應（4 篇）

這些論文的 % 來自改變「環境溫度」而非自熱，不可當 SHE 劣化基準引用（最常見的分類陷阱，例如 Rathore 2022 的 -8.2%）。

論文 / IEEE / 年 / DOI	元件	其 % 數字（非 SHE）	偏壓	查證
Design Optimization of Three-Stacked Nanosheet FET From Self-Heating E... (IEEE, 2022) DOI: 10.1109/tdmr.2022.3181672	Three-stacked Si nanosheet FET; partially-depleted silicon...	Abstract states SHE 'raises lattice temperature several degrees above ambient and degrades the driving current...	not accessed	confirmed
Ambient Temperature-Induced Device Self-Heating Effects on Multi-Fin S... (IEEE, 2020) DOI: 10.1109/ted.2020.2975416	Multi-fin Si FinFET CMOS inverter / ring oscillator, N-14 ...	No device-level Ion % in abstract (confirmed); circuit-level degradation (delay, noise margin, gain) quantifie...	not accessed (closed access)	confirmed
Ambient Temperature-Induced Device Self-Heating Effects on Multi-Fin S... (IEEE, 2018) DOI: 10.1109/ted.2018.2834979	Si 3-fin bulk n-FinFET, sub-14-nm technology node	No % in abstract (confirmed); performance-metric degradation vs TA and TCR is figure-level only	not accessed (closed access)	confirmed
Study on the regulation factors and mechanism of self-heating effects ... (非, 2024) DOI: 10.1088/1361-6641/ad689f	14 nm bulk FinFET,圓柱梯形(非矩形)fin;FinAngle 84°-90°掃描	TA 300->400K:驅動電流僅 -0.21%(2.33051e-5->2.3257e-5 A);TA 220K 時 +0.87%。此為 ta_sweep,不得計入 SHE%;論文無 electrothermal-vs-...	Vgs=Vds=0.8 V;TA=300-400 K(另有 220 K 點)	confirmed

資料來源：多代理文獻獵取工作流（loop-until-dry，第 1-2 輪完成）-> 逐篇對抗式查證（confirmed 105、value_corrected 6、reclassified 6、plausible_unverified 4；0 篇 likely_hallucinated）。第 3 輪追加獵取因帳號 session 上限中止，邊際新增有限；本目錄為已查證 121 篇之完整彙整。標「圖層級/paywall/not accessed」之數字，引用前務必自讀原文；標「reclassified」者多為原被誤標成 SHE 的 ta_sweep 已更正。